

Comprehensive Engineering Study Notes: Fundamentals of Capacitance & RC Transients

1. High-Level Overview & Intuitive Foundations

What is Capacitance?

In electrical engineering, a **capacitor** is a passive electrical component designed to store energy in an **electric field**. Unlike a battery, which generates electrical energy through internal chemical reactions, a capacitor merely holds electrical charge physically on conductive plates separated by an insulating medium (dielectric).

- Core Intuition:** Imagine a capacitor as an elastic water storage balloon in a pipe system. When water (current) flows in, the balloon expands, building back-pressure (voltage). The balloon resists rapid changes in water level.
- Fundamental Law:** Capacitance (C) is the fundamental property of a capacitor that **opposes any sudden rate of change of voltage** across its terminals.

KEY CONCEPT	
	Voltage across a capacitor CANNOT change instantaneously.
	To change voltage instantly would require an infinite amount of current,
	which is physically impossible in real circuits: $i_c = C \frac{dv_c}{dt}$.

Governing Mathematical Equations

1. Stored Energy:

The energy E stored within the electrostatic field of a capacitor is directly proportional to its capacitance and the square of the voltage across it:

$$E = \frac{1}{2}CV^2 \quad [\text{Joules, J}]$$

2. Current-Voltage Relationship:

Current through a capacitor depends on how quickly the voltage across it changes over time (the derivative of voltage with respect to time):

$$i_c(t) = C \frac{dv_c(t)}{dt} \quad [\text{Amperes, A}]$$

3. Voltage as an Integral of Current:

By integrating the current equation, the instantaneous voltage across the capacitor is accumulated charges over time:

$$v_c(t) = \frac{1}{C} \int i_c(t) dt \quad [\text{Volts, V}]$$

2. Transient Behavior in RC Series Circuits

What is a "Transient" State?

When a switch changes position in a circuit containing capacitors or inductors, the circuit undergoes a transition period before settling into a steady condition.

- Transient State:** The temporary response of a circuit during the transition phase right after a disturbance (e.g., closing/opening a switch).
- Steady State:** The long-term behavior of the circuit after all voltages and currents have settled into fixed or periodic patterns ($t \rightarrow \infty$).

Mathematical Derivation of Circuit Differential Equation

Consider a series RC circuit connected to a constant DC voltage source V via a switch that closes at time $t = 0$.

 Diagram: Series RC Circuit with DC Voltage Source

Visual Layout Description: A simple single-mesh loop containing a DC voltage source V on the left branch, connected in series with a single-pole single-throw switch that closes at $t = 0$, followed by a resistor R on the top branch, and a capacitor C on the right branch. The current $i_C(t)$ flows clockwise around the loop, and voltage polarity across C is marked positive at the top terminal and negative at the bottom terminal as $v_C(t)$.

Step 1: Apply Kirchhoff's Voltage Law (KVL)

Applying KVL around the loop for $t \geq 0$:

$$V = v_R(t) + v_C(t)$$

Using Ohm's Law for the resistor, $v_R(t) = R \cdot i_C(t)$:

$$V = Ri_C(t) + v_C(t)$$

Step 2: Substitute $i_C(t)$ into KVL

Substitute $i_C(t) = C \frac{dv_C(t)}{dt}$:

$$V = RC \frac{dv_C(t)}{dt} + v_C(t)$$

This is a first-order linear non-homogeneous differential equation.

Step 3: Solve the First-Order Differential Equation

The total solution $v_C(t)$ is composed of two parts:

$$v_C(t) = v_{c.f.}(t) + v_{p.i.}(t)$$

- **Complementary Function ($v_{c.f.}$):** Represents the **Transient Response** (natural response).

$$v_{c.f.}(t) = K_1 e^{-t/RC}$$

- **Particular Integral ($v_{p.i.}$):** Represents the **Steady-State Response** (forced response).

$$v_{p.i.}(t) = V$$

Combining both terms gives the general expression:

$$v_C(t) = K_1 e^{-t/RC} + V$$

Step 4: Apply Initial Boundary Conditions

Assuming the capacitor is completely uncharged before the switch closes at $t = 0^-$:

$$v_C(0^+) = v_C(0^-) = 0$$

Substitute $t = 0$ into the general equation:

$$0 = K_1 e^0 + V \implies K_1 = -V$$

Step 5: Final Expressions for Voltage and Current

1. **Capacitor Charging Voltage Equation:**

$$v_C(t) = V \left(1 - e^{-t/RC} \right)$$

2. **Charging Current Equation:**

$$i_c(t) = C \frac{dv_c}{dt} = C \cdot V \left(0 - \left(-\frac{1}{RC} \right) e^{-t/RC} \right)$$

$$i_c(t) = \frac{V}{R} e^{-t/RC}$$

The Time Constant (τ)

The quantity $\tau = RC$ is defined as the **Time Constant** of the circuit (measured in seconds).

$$\tau = R \times C \quad [\text{Seconds, s}]$$

- **Physical Meaning:** τ dictates how rapidly the circuit responds.
- At $t = \tau$:

$$v_c(\tau) = V (1 - e^{-1}) \approx V(1 - 0.368) = 0.632V$$

The time constant τ is the exact time required for the capacitor voltage to reach approximately **63.2%** of its maximum steady-state voltage.

- **Practical Rule of Thumb:** A circuit is considered to have reached steady state after 5τ (at $t = 5\tau$, $v_c \approx 99.3\%$ of V).

 Diagram: RC Charging Curve Graph

Visual Layout Description: A 2D Cartesian plot showing time t on the horizontal axis and capacitor voltage $v_c(t)$ on the vertical axis. The curve starts at the origin $(0, 0)$ and curves upward smoothly, asymptotically approaching the maximum steady-state DC voltage $V = 10 \text{ V}$. A dashed tangent line from the origin reaches $V = 10 \text{ V}$ at time $t = \tau = 0.01 \text{ s}$, illustrating the initial rate of rise.

3. Step-by-Step Worked Examples & Problems

Problem 1: Full RC Circuit Parameter Analysis

Problem Statement:

A capacitor of $50 \mu\text{F}$ in series with a $1 \text{ k}\Omega$ resistor is connected across a 100 V DC supply. Calculate:

1. Time constant (τ) of the charging circuit
2. Initial charging current (I_0)
3. Equation of current as a function of time ($i(t)$)
4. Value of current after 0.05 s
5. Charging current after 0.06 s
6. Equation of charge as a function of time ($q(t)$)
7. Voltage across the resistor after 0.06 s

Detailed Step-by-Step Solution

Given Data:

- $C = 50 \mu\text{F} = 50 \times 10^{-6} \text{ F}$
- $R = 1 \text{ k}\Omega = 1000 \Omega$
- $V = 100 \text{ V}$

Part 1: Calculate Time Constant (τ)

$$\tau = R \cdot C = (1000 \, \Omega) \times (50 \times 10^{-6} \, \text{F}) = 0.05 \, \text{s}$$

Part 2: Calculate Initial Charging Current (I_0)

At $t = 0$, $v_c(0) = 0 \, \text{V}$. The entire supply voltage drops across R :

$$I_0 = \frac{V}{R} = \frac{100 \, \text{V}}{1000 \, \Omega} = 0.1 \, \text{A} = 100 \, \text{mA}$$

Part 3: Derive Equation of Current as a Function of Time $i(t)$

Using $i(t) = \frac{V}{R} e^{-t/\tau}$:

$$i(t) = 0.1 e^{-t/0.05} = 0.1 e^{-20t} \quad [\text{A}]$$

Part 4: Calculate Current at $t = 0.05 \, \text{s}$

Note that $t = 0.05 \, \text{s} = 1\tau$:

$$i(0.05) = 0.1 \cdot e^{-0.05/0.05} = 0.1 \cdot e^{-1} = 0.1 \times 0.367879 = 0.03679 \, \text{A} = 36.79 \, \text{mA}$$

Part 5: Calculate Current at $t = 0.06 \, \text{s}$

$$i(0.06) = 0.1 \cdot e^{-0.06/0.05} = 0.1 \cdot e^{-1.2} = 0.1 \times 0.30119 = 0.03012 \, \text{A} = 30.12 \, \text{mA}$$

Part 6: Derive Equation of Charge as a Function of Time $q(t)$

Since $q(t) = C \cdot v_c(t)$ and $v_c(t) = V(1 - e^{-t/\tau})$:

$$q(t) = CV(1 - e^{-t/\tau})$$

$$Q_{\max} = CV = (50 \times 10^{-6} \, \text{F}) \times (100 \, \text{V}) = 5 \times 10^{-3} \, \text{C} = 5 \, \text{mC}$$

$$q(t) = 5 \times 10^{-3} (1 - e^{-20t}) \quad [\text{C}]$$

Part 7: Calculate Voltage Across Resistor at $t = 0.06 \, \text{s}$

Using Ohm's Law $v_R(t) = i(t) \cdot R$:

$$v_R(0.06) = (0.03012 \, \text{A}) \times (1000 \, \Omega) = 30.12 \, \text{V}$$

(Alternatively: $v_c(0.06) = 100(1 - e^{-1.2}) = 69.88 \, \text{V} \implies v_R = 100 - 69.88 = 30.12 \, \text{V}$)

Problem 2: Current-Driven RC Circuit & Target Voltage Time Computation

Problem Statement:

In the circuit shown below, switch S is closed to position 1 at time $t = 0$. How long does it take for the capacitor to attain 70% of its final voltage? Assume the capacitor is initially uncharged. Find the time constant (τ) of the circuit after the switch is closed.

Diagram: Current Source Driven RC Circuit Switch Diagram

Visual Layout Description: A multi-mesh circuit fed by a 1 A ideal current source on the far left branch. A single-pole switch S at node 1 closes at $t = 0$. Branch 1 contains a $200\ \Omega$ vertical resistor connected directly across the node after the switch. To the right, a horizontal $100\ \Omega$ resistor connects to node a . From node a to bottom node b , a 0.01 F capacitor is connected in parallel with a vertical $100\ \Omega$ resistor.

Step-by-Step Solution

Step 1: Simplify Circuit using Thévenin's Equivalent as Seen by Capacitor

To simplify the network connected to capacitor $C = 0.01\text{ F}$, replace everything to the left of terminals a - b with its Thévenin Equivalent (V_{th} and R_{th}).

1. Find Open-Circuit Voltage (V_{th}):

Disconnect the capacitor at terminals a - b .

The 1 A current source flows through the parallel branch of the $200\ \Omega$ resistor and the branch containing $(100\ \Omega + 100\ \Omega)$:

- Branch 1 Resistance: $200\ \Omega$
- Branch 2 Resistance: $100\ \Omega + 100\ \Omega = 200\ \Omega$

By current division, current through the right branch:

$$I_{right} = 1\text{ A} \times \frac{200}{200 + 200} = 0.5\text{ A}$$

The open-circuit voltage across terminals a - b (across the rightmost $100\ \Omega$ resistor):

$$V_{th} = I_{right} \times 100\ \Omega = 0.5\text{ A} \times 100\ \Omega = 50\text{ V}$$

2. Find Thévenin Resistance (R_{th}):

Deactivate the independent current source (open-circuit it). Looking back into terminals a - b :

- The $200\ \Omega$ resistor is in series with the top $100\ \Omega$ resistor $\implies 200 + 100 = 300\ \Omega$.
- This $300\ \Omega$ combination is in parallel with the rightmost $100\ \Omega$ resistor.

$$R_{th} = (200 + 100) \parallel 100 = 300 \parallel 100 = \frac{300 \times 100}{300 + 100} = \frac{30000}{400} = 75\ \Omega$$

Step 2: Calculate Equivalent Circuit Time Constant (τ)

Using Thévenin equivalent parameters:

$$\tau = R_{th} \cdot C = 75\ \Omega \times 0.01\text{ F} = 0.75\text{ s}$$

Step 3: Compute Time to Reach 70% of Final Voltage

The target voltage is $v_c(t) = 0.70 \cdot V_{th}$.

Using the standard charging equation:

$$v_c(t) = V_{th} \left(1 - e^{-t/\tau}\right)$$

$$0.70 \cdot V_{th} = V_{th} \left(1 - e^{-t/0.75}\right)$$

$$0.70 = 1 - e^{-t/0.75}$$

$$e^{-t/0.75} = 0.30$$

Taking the natural logarithm ($\ln$) on both sides:

$$-\frac{t}{0.75} = \ln(0.30)$$

$$-\frac{t}{0.75} = -1.20397$$

$$t = 1.20397 \times 0.75 \approx 0.903 \text{ s}$$

Problem 3: Two-Position Switch Transition Analysis

Problem Statement:

In the circuit shown below, the switch has been in **position A** for a long time. At $t = 0$, the switch moves to **position B**.

1. Determine $v(t)$ for $t > 0$.
2. Calculate its value at $t = 1 \text{ s}$ and $t = 4 \text{ s}$.

Diagram: Two Position Switch RC Circuit Analysis

Visual Layout Description: A multi-mesh circuit with two sub-circuits selectable via a single-pole double-throw switch. Left sub-circuit (Position A) contains a 24 V source in series with a $3 \text{ k}\Omega$ resistor, connected in parallel across a $5 \text{ k}\Omega$ resistor. Position B connects to a right sub-circuit containing a $4 \text{ k}\Omega$ resistor in series with a 30 V source. A 0.5 mF capacitor is connected between the central switch wiper and ground.

Step-by-Step Solution

Phase 1: Determine Initial Condition at $t = 0^-$ (Switch in Position A)

Before $t = 0$, the switch was at position A for a "long time" ($t \rightarrow \infty$). Thus, the capacitor acts as an **open circuit** (fully charged).

Calculate the voltage across the $5 \text{ k}\Omega$ resistor using the voltage divider rule:

$$v_c(0^-) = 24 \text{ V} \times \frac{5 \text{ k}\Omega}{3 \text{ k}\Omega + 5 \text{ k}\Omega} = 24 \times \frac{5}{8} = 15 \text{ V}$$

Since capacitor voltage cannot change instantaneously:

$$v(0^+) = v(0^-) = 15 \text{ V}$$

Phase 2: Determine Final Steady-State Condition as $t \rightarrow \infty$ (Switch in Position B)

At $t = 0$, the switch flips to position B. The left circuit is disconnected. The capacitor is now connected to the right branch containing $R = 4 \text{ k}\Omega$ and source $V = 30 \text{ V}$.

After a long time ($t \rightarrow \infty$), the capacitor charges to the new DC source voltage:

$$v(\infty) = 30 \text{ V}$$

Phase 3: Find Circuit Time Constant τ for $t > 0$

Looking from the capacitor terminals into Position B:

$$R_{eq} = 4\text{ k}\Omega = 4000\ \Omega$$

$$C = 0.5\text{ mF} = 0.5 \times 10^{-3}\text{ F} = 500\ \mu\text{F}$$

$$\tau = R_{eq} \cdot C = 4000 \times (0.5 \times 10^{-3}) = 2.0\text{ s}$$

Phase 4: Construct General Transient Equation for $v(t)$

The general expression for any first-order transient response is:

$$v(t) = v(\infty) + [v(0^+) - v(\infty)]e^{-t/\tau}$$

Substitute the values:

- $v(0^+) = 15\text{ V}$
- $v(\infty) = 30\text{ V}$
- $\tau = 2.0\text{ s}$

$$v(t) = 30 + (15 - 30)e^{-t/2.0}$$
$$v(t) = 30 - 15e^{-0.5t} \quad [\text{V}] \quad \text{for } t \geq 0$$

Phase 5: Calculate Voltage at Specified Time Instants

1. At $t = 1\text{ s}$:

$$v(1) = 30 - 15e^{-0.5(1)} = 30 - 15e^{-0.5} = 30 - 15(0.60653) = 30 - 9.098 = 20.902\text{ V}$$

2. At $t = 4\text{ s}$:

$$v(4) = 30 - 15e^{-0.5(4)} = 30 - 15e^{-2.0} = 30 - 15(0.135335) = 30 - 2.030 = 27.970\text{ V}$$

4. Chapter Core Summary Table

Parameter / Concept	Symbol / Equation	Physical Unit	Key Engineering Concept
Capacitance	$C = \frac{Q}{V}$	Farads (F)	Opposition to rapid changes in voltage.
Stored Field Energy	$E = \frac{1}{2}CV^2$	Joules (J)	Energy stored in the internal electrostatic field.
Capacitive Current	$i_c(t) = C\frac{dv_c}{dt}$	Amperes (A)	Proportional to the derivative (rate of change) of voltage.
Time Constant	$\tau = RC$	Seconds (s)	Time taken to reach 63.2% of final step change.
Charging Voltage	$v_c(t) = V(1 - e^{-t/\tau})$	Volts (V)	Exponential growth from initial voltage to source voltage.
Discharging Voltage	$v_c(t) = V_0e^{-t/\tau}$	Volts (V)	Exponential decay of voltage when source is removed.
General Response	$v(t) = v(\infty) + [v(0^+) - v(\infty)]e^{-t/\tau}$	Volts (V)	Universal formula applicable to all 1st-order RC transients.